

Chapter 113 — Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

COMBINATION THERAPY

Antimuscarinics and β 3-Agonists

- **Different mechanisms of action** → rationale for combining an antimuscarinic with a β 3-adrenoceptor agonist in overactive bladder (OAB) and detrusor overactivity (DO), particularly when the chosen antimuscarinic dose gives **intolerable adverse events (AEs)** or **insufficient symptom improvement**.
- Solifenacin 10 mg plus mirabegron 50 mg reduced urgency episodes more than solifenacin 5 mg; the combination **roughly doubled constipation** against solifenacin 10 mg alone, with **no difference in heart rate, blood pressure or electrocardiogram**.

COMPARISON: ANTIMUSCARINIC–MIRABEGRON COMBINATION TRIALS

Trial	Design	Population and arms	Result
SYNERGY	Phase III, 12 weeks, six arms	Wet OAB; placebo; solifenacin 5 mg + mirabegron 25 or 50 mg; solifenacin 5, mirabegron 25 or 50 mg alone	Co-primary endpoints: daily incontinence (UI) episodes and voids. Solifenacin 5 + mirabegron 50 → greater fall in daily UI than placebo and either monotherapy; better quality of life (QoL): OAB Questionnaire (OAB-q) Symptom Bother, health-related QoL (HRQoL) total, Patient Perception of Bladder Condition (PPBC)
SYNERGY II	Phase III, 12 months	Wet OAB; solifenacin 5 + mirabegron 50 vs each monotherapy	Superior for UI episodes and micturitions; dry mouth commonest AE, similar across arms → long-term safety
Network meta-analysis	—	Solifenacin 5 + mirabegron 50 vs mirabegron 50	More effective for reducing UI and achieving full continence
BESIDE	Phase IIIb; 4-week solifenacin 5 run-in, then 12 weeks	Solifenacin 5 non-responders; + mirabegron (25 → 50 mg) vs solifenacin 5 vs solifenacin 10	Superior to solifenacin 5 for daily UI (primary) but not to solifenacin 10 . Continence 46.0% vs 37.9% (5 mg) vs 40.2% (10 mg). Dry mouth highest with solifenacin 10; combination ≈ solifenacin 5
MILAI	Phase IV, Japan, 16 weeks	Insufficient response to solifenacin 2.5 or 5 mg for ≥4 weeks; + mirabegron 25 mg, → 50 mg after 8 weeks if needed	No serious AEs; Overactive Bladder Symptom Score (OABSS), OAB-q short form (SF) and bladder diary improved

Trial	Design	Population and arms	Result
MILAI II	52 weeks; first long-term antimuscarinic add-on study	Insufficient response after 6 weeks of mirabegron 50 mg; + solifenacin 5, propiverine 20, imidafenacin 0.2 mg or tolterodine extended release (ER) 4 mg	Dose could be doubled after 8 weeks (not tolterodine ER), reduced if AEs. All arms improved, sustained to 52 weeks; AEs similar across arms (dry mouth, nasopharyngitis, constipation commonest); no concern on serious AEs, vital signs, QT interval or post-void residual (PVR)

- **What the table tests:** the direction of add-on. The combination beats **solifenacin 5 mg** and each monotherapy, but in non-responders it matches, rather than beats, **dose-escalated solifenacin 10 mg**, with less dry mouth → it is an **alternative to dose escalation**.
- **BESIDE subanalyses:** benefit **irrespective of age** (≥ 65 , ≥ 75 years); responders (≥ 10 -point OAB-q Symptom Bother and HRQoL, ≥ 2 -point PPBC) favour combination.
- **Trap:** assuming any antimuscarinic combines with mirabegron on the same evidence — mirabegron add-on is tested **only on solifenacin** (see pair below).

DIRECTION OF ADD-ON

Mirabegron added to an antimuscarinic	Antimuscarinic added to mirabegron
Antimuscarinic first line, response inadequate	Mirabegron first line, response inadequate
SYNERGY, SYNERGY II, BESIDE, MILAI	MILAI II
Solifenacin only	Four antimuscarinics: solifenacin, propiverine, imidafenacin, tolterodine ER

Antimuscarinics With Vibegron

- **Benefit extrapolated from mirabegron;** direct evidence limited, validation awaited.
- **Two-part phase IIb study:** part 1 → vibegron 3, 15, 50 or 100 mg, tolterodine ER 4 mg (positive control) or placebo for 8 weeks, or vibegron/tolterodine 50/4 mg for 4 weeks → vibegron 50 mg for 4 weeks; part 2 → vibegron 100 mg, tolterodine 4 mg, the 100/4 mg combination or placebo for 4 weeks.
- **Vibegron/tolterodine 100/4 mg** → greater fall in daily micturitions and urgency than either monotherapy; **numerically** greater fall in urgency incontinence (UUI) in wet OAB.

KEY POINTS: COMBINATION OF ANTIMUSCARINICS AND B3-AGONISTS

- **Different mechanisms** → combination is useful in OAB/DO, especially with intolerable AEs or insufficient response to the antimuscarinic.
- **Solifenacin fails** → **add mirabegron:** efficacy and safety well documented; other antimuscarinics with mirabegron not investigated.
- **Mirabegron first line fails** → **add an antimuscarinic:** effective and safe (solifenacin 5 mg, propiverine 20 mg, imidafenacin 0.2 mg, tolterodine ER 4 mg).
- **No trial** of antimuscarinic–vibegron coadministration after failure of either monotherapy.

Concept — The evidence is **directional and drug-specific:** *mirabegron on solifenacin and an antimuscarinic on mirabegron are each proven, and nothing else is. Against a dose-escalated antimuscarinic, the combination gains tolerability, not efficacy.*

α-Blockers and Desmopressin

- **α-blocker plus desmopressin** → consider for **nocturia resistant to first-line treatment**.
- **About 30% of men aged ≥60 years with benign prostatic hyperplasia (BPH) have nocturnal polyuria (NP)** (phase IV study), a cause of nocturia that α-blockade does not address.
- **α-blocker-resistant BPH**: oral desmopressin → most patients achieve a **≥50% fall in nocturnal voids** (two non-placebo-controlled studies).
- **Low-dose oral desmopressin 0.1 mg** in men with BPH aged ≥65 years, over 12 months → fewer nocturnal voids and lower nocturnal urine volume.
- **No placebo control** → placebo contribution unknown.

Vaginal Oestrogens and Anticholinergics

COMPARISON: VAGINAL OESTROGEN WITH ANTIMUSCARINIC THERAPY

Comparison	Population	Result
Tolterodine ± vaginal oestrogen cream	Postmenopausal OAB	Combination superior
Tolterodine added after 12 weeks of intravaginal estradiol, vs the reverse order	OAB	Greater improvement; not the inverse
Tolterodine ± topical oestrogen	Symptomatic DO	No difference
Solifenacin 5 mg ± vaginal oestrogen capsules (randomised controlled trial, RCT)	Postmenopausal OAB	No significant difference in voids/24 h (primary) or urgency
Estradiol vaginal ring vs oral oxybutynin, 12 weeks	OAB	No difference
Fesoterodine ± topical vaginal oestrogen (small)	OAB	No significant difference at 12 weeks; numerical trend to combination

- **Pattern**: positive signals are weak and **sequence-dependent** (antimuscarinic after oestrogen); simultaneous-combination RCTs are null; vaginal estradiol alone ≈ **oxybutynin**.
- **Trap**: treating additive benefit of oestrogen plus antimuscarinic as established.

Botulinum Toxin A and Antimuscarinics

- **Neurogenic DO (NDO)** from spinal cord injury (SCI) or multiple sclerosis (MS). In the pooled **DIGNITY** studies, onabotulinumtoxinA (onabotA) **200 U** reduced UI **equally in anticholinergic users and non-users**, with:
 - identical proportions achieving 50% and 100% UI reduction
 - similar rises in maximum cystometric capacity and falls in maximum detrusor pressure
 - similar Incontinence Quality of Life (I-QOL) scores.
- **Unanswered**: whether prior partial anticholinergic responders gain from combination.
- **EMBARK** (idiopathic OAB, onabotA 100 U) enrolled only anticholinergic-refractory patients → **no user vs non-user comparison**.
- **In practice**, anticholinergics or β3-agonists are likely used to prolong the **tail of onabotA 100 U** — unevaluated.

	DIGNITY	EMBARK
Population	NDO (SCI, MS)	Idiopathic OAB, anticholinergic-refractory
OnabotA dose	200 U	100 U
Users vs non-users	Equal efficacy	Not comparable

Concept — OnabotA efficacy is independent of concurrent anticholinergic use in NDO. Whether the combination adds benefit has never been tested.

DRUGS THAT FAILED IN CLINICAL TRIALS TO IMPROVE STORAGE OR VOIDING FUNCTION

- **Scope:** drugs that **failed in large clinical trials or showed no efficacy in pilot studies**. Compounds tested only in animals are excluded. Future drug design may still rescue some of these classes.

COMPARISON: FAILED AGENTS FOR STORAGE DYSFUNCTION

Agent	Mechanism / route	Indication; phase	Outcome and reason for failure
Vitamin D3 supplement	Vitamin D3 receptor	UUI; pilot	↓ UUI vs placebo; frequency, nocturia and urgency unchanged; obesity blunts effect
Elocalcitol 75 or 150 µg	Synthetic vitamin D3 analogue	Women with OAB and idiopathic DO; phase IIb, 4 weeks	Volume at first involuntary contraction unchanged ; ↓ incontinence, ↑ PPBC; well tolerated; clinical data limited
Verapamil	L-type Ca ²⁺ channel blocker, intravesical	Urgency or UUI; urodynamic study vs oxybutynin, trospium, placebo	No relevant change in bladder function
Nimodipine 30 mg/day	Dihydropyridine Ca ²⁺ channel blocker	Elderly UUI; small RCT	Not superior to placebo
ZD0947 25 mg/day	adenosine triphosphate (ATP)-sensitive K ⁺ channel opener	OAB; phase II	No relevant gain in mean voided volume or OAB symptoms
Flupirtine 400 → 600 mg/day	KCNQ (Kv7) K ⁺ channel modulator; central non-opioid analgesic	OAB; phase II, 12 weeks, vs placebo and tolterodine	Halted for hepatotoxicity ; better than placebo but worse than tolterodine for frequency; inconclusive
URO-902	Naked plasmid DNA encoding BK channel α subunit (gene therapy); bladder injection or instillation	Women with OAB; phase 1	No AE withdrawals. Injection: ↓ urgency and voids at 1 week. Instillation: trend only (↓ involuntary contractions, UUI). No further studies
PSD503	Phenylephrine 20% vaginal gel, α-adrenoceptor (α-AR) agonist	Stress urinary incontinence (SUI); small placebo-controlled trial	↓ pad weight gain vs placebo; poor patient acceptance ; abandoned

Agent	Mechanism / route	Indication; phase	Outcome and reason for failure
Phenylpropanolamine, midodrine, norepinephrine	Systemic α -AR agonists	SUI	Evidence very weak or absent. Non-selective → hypertension, haemorrhagic stroke , sleep disturbance, headache, tremor, palpitations. US Food and Drug Administration (FDA) ban, December 2003
Clenbuterol 20 μ g twice daily	β -AR agonist	Women with SUI; RCT, 2 weeks	Positive, but baseline pad weights unbalanced . High dry rates in mild SUI, alone or with pelvic floor exercises. Finger tremor, tachycardia; no further studies
RimabotulinumtoxinB	Botulinum toxin B (BoNT/B)	DO refractory to toxin A	Short duration : ≤ 3 months at highest dose (dry mouth, malaise); ≤ 10 weeks at 5,000 U

- **Pattern**: positive signals are confined to **pilot, phase I–II or animal data**; none was confirmed in a large trial. Failure modes: no clinical effect, **toxicity** (flupirtine, systemic α -agonists), **short duration** (BoNT/B), **acceptability** (PSD503) or **flawed design** (clenbuterol).
- **Trap**: the α -agonist gel reduced leakage and still failed, on **acceptability**, not efficacy.

Vitamin D3 Receptor Agonists

- **Inverse association between vitamin D3 and LUTS** (lower urinary tract symptoms): higher levels $\leftrightarrow$ higher urinary flow; lower levels $\leftrightarrow$ UI, **independent of ethnicity and age**. In men, low vitamin D $\leftrightarrow$ higher OABSS **in winter only**.
- **Elocalcitol** (formerly BXL-628): no effect on Rho kinase expression; delayed bladder-strip contraction; increased L-type Ca^{2+} channel expression (potentially enhancing contraction).
 - **Animal models**: improved bladder function after acetic acid, bladder outlet obstruction (BOO) and cyclophosphamide cystitis.
 - **Maxacalcitol** reduced retinyl-acetate-induced DO.

Calcium Antagonists

- **Ca^{2+} channels set smooth muscle tone** via free intracellular calcium. **Voltage-gated channels** (Q- and L-type), not store-operated channels, are implicated in the bladder.
- **L-type evidence**: inhibitors block human bladder contraction in vitro; knockout mice have impaired contractility.
- **Clinical paradox**: despite decades of cardiovascular use, **no major reports of impaired bladder contractility** → no clinical evidence for calcium channel blockers in storage.

Potassium Channel Openers

- **K^+ channel opening → hyperpolarisation → smooth muscle relaxation**; many K^+ channel types regulate bladder tone.
- **K^+ channels also regulate cardiomyocytes** → the class has probably been **tested at excessively low doses** in OAB to avoid cardiovascular AEs.
- **Flupirtine data** support developing **KCNQ agents without liver toxicity**.
- **Big potassium (BK, maxi-K) channel**, large-conductance and voltage- and calcium-activated, is **highly expressed on bladder smooth muscle** → target of gene therapy (URO-902).

α -AR and β -AR Agonists

- **α -AR agonists** aim to **increase urethral closing power** in SUI; systemic agents **lack urethral α -AR selectivity** → the cardiovascular toxicity in the table. **β -AR agonists**: positive studies never followed up.

Botulinum Toxin Subtype B

- **Dose escalation** 2,500–15,000 U at 10 bladder sites (women with OAB): longest effect only at the highest, poorly tolerated doses → **short duration at safe doses** limits usefulness.

Concept — **Failure is mostly a problem of selectivity or therapeutic window, not a lack of bladder effect.**

Calcium antagonists, K^+ channel openers and α -agonists all act on bladder or urethra, but the same targets in heart and vessels cap the tolerable dose.

DRUGS THAT FAILED TO IMPROVE VOIDING DYSFUNCTION

Toxins Injected into the Prostate

COMPARISON: INTRAPROSTATIC TOXINS FOR BPH-RELATED LUTS

Agent	Mechanism	Route; phase	Result; fate
OnabotA 100–300 U	Apoptosis of prostatic epithelium and stroma (rat, dog)	Ultrasound-guided, transition zone, transrectal or transperineal; phase II, then two large RCTs	Small studies: ↓ volume, ↓ LUTS, no sexual AEs. Phase II: 30% improvement in American Urological Association Symptom Index (AUA-SI), some urosepsis. Large RCTs: = saline , no dose response
PRX302 0.6 µg/g prostate	Modified proaerolysin : pore-forming bacterial protoxin activated by prostate-specific antigen (PSA)	Transperineal ultrasound-guided, transition zone; phase IIb, 2:1 vs vehicle (2% human albumin)	At 3 months: International Prostate Symptom Score (IPSS) ↓ about 9 points, peak flow (Q _{max}) ↑ 3 mL/s, significant vs vehicle, but large placebo effect . Mild transient local pain only; no storage or sexual AEs. Not developed further
NX-1207	Proprietary protein, selective pro-apoptotic	Single transition-zone injection; phase II → III	Phase II allegedly > finasteride (never peer-reviewed); phase III negative → suspended

- **OnabotA RCT entry criteria**: IPSS >12 with Q_{max} 5–15 mL/s (dose-ranging, transperineal or transrectal); IPSS >14 with Q_{max} 4–15 mL/s (200 U vs saline, transrectal).
- **PRX302 entry criteria**: IPSS ≥15, Q_{max} ≤12 mL/s, prostate 30–100 mL. Washout: α -blockers ≥4 weeks, 5 α -reductase inhibitors (5ARIs) ≥6 months.
- **Later small studies** favouring onabotA over optimised medical therapy, or over transurethral resection of the prostate (TURP) in drug-refractory men, are **unlikely to restart investigation**.
- **Trap**: taking uncontrolled or phase II improvement as efficacy. Each toxin's LUTS benefit shrank or vanished against **sham injection**.

Concept — **Prostatic injection carries a large placebo effect of its own.** *Every toxin looked effective until it was compared with a sham injection, which is why none survived.*